

Release of poly- and perfluoroalkyl substances from finished biosolids in soil mesocosms

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ABSTRACT

Finished biosolids were collected and characterized from seven municipal water resource recovery facilities. Poly- and perfluoroalkyl substances (PFAS) for the 54 quantified in the biosolids ranged from 323 ± 14.1 to 1100 ± 43.8 $\mu\text{g/kg}$ (dry weight basis). For all biosolids, greater than 75% of the PFAS fluorine mass was associated with precursors. Di-substituted polyfluorinated phosphate esters (diPAPs) were the most abundant PFAS identified in the biosolids. The total oxidizable precursor assay on biosolids extracts generally failed to quantify the amount of precursors present, in large part due to the fact that diPAPs were not fully transformed during the TOP assay. Outdoor biosolids column leaching experiments intended to simulate biosolids land application showed sustained PFAS leaching over the 6-month study duration. Perfluoroalkyl acid (PFAA) concentrations in leachate, when detected, typically ranged in the 10 s to 100 s of ng/L ; no diPAPs were detected in the leachate. The PFAA leaching from the biosolids exceeded the PFAA mass initially present in the biosolids (typically by greater than an order of magnitude), but the cumulative PFAA mass leached did not exceed the molar equivalents that could be explained by transformation of quantified precursors. These results highlight the importance of PFAA precursors initially present in biosolids and their contribution to long term leaching of PFAAs from land-applied biosolids.

1. Introduction

Public and regulatory focus on poly- and perfluoroalkyl substances (PFAS) in the environment has increased in recent years as evidence regarding the prevalence and persistence of these compounds in the environment has emerged. The wide-spread use of PFAS in consumer products ranging from non-stick surfaces, to water and grease repelling textiles, to aqueous film forming foams (AFFFs) for firefighting have resulted in PFAS detections in the influent and effluent of many water resource recovery facilities (WRRFs) (Rahman et al., 2014; Hu et al., 2016; Ahrens et al., 2009), along with detections in sludge and biosolids (Ebrahimi et al., 2021; Sepulvado et al., 2011).

A United States national survey of 13 short- and long-chain perfluoroalkyl acids (PFAAs) in biosolids from 94 WRRFs in 2001 showed that PFOS was the most abundant PFAA detected in biosolids, at a concentration of 403 ± 127 $\mu\text{g/kg}$ of dry weight, followed by PFOA at 34

± 22 $\mu\text{g/kg}$; remaining reported PFAAs ranged from 2 to 26 $\mu\text{g/kg}$ (Venkatesan and Halden, 2013). Analysis of 12 PFAAs in biosolids from a highly urban US WRRF that were collected quarterly from 2005 to 2013 showed that PFNA, PFOA, and PFOS had the highest average concentrations (25.1 $\mu\text{g/kg}$, 23.5 $\mu\text{g/kg}$, and 22.5 $\mu\text{g/kg}$, respectively); the remaining analytes were less than 8 $\mu\text{g/kg}$ (Armstrong et al., 2016). An Australian study examining biosolids collected from 19 WRRFs in 2018 showed, based on a PFAS analyte list consisting of 44 compounds, a total PFAS concentration of 260 $\mu\text{g/kg}$ dry weight (Moodie et al., 2021).

Measured increases in PFAS levels due to precursor transformation during wastewater treatment in WRRFs also have been observed (Schultz et al., 2006; Chen et al., 2018; Lakshminarasimman et al., 2021). With respect to final biosolids, increases in PFAA concentrations due to precursor transformation were observed during pelletization (19% increase) and alkaline stabilization (99% increase)

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Table 1

Solids processing for the seven biosolids used in this study.

Unique Identifier	Thickening	Digestion	Dewatering	Drying	Finished Biosolids Class ¹	Solids Type
A1	Gravity thickeners	None	Centrifuge and Press	Yes	A	Pellets
A2	Gravity thickeners	M	Settling	No	U	Sludge ²
B1	Centrifuges	M	Centrifuge	No	B	Cake
C1	Gravity and centrifugal thickeners	M	Centrifuge	Yes	A	Pellets
D1	Gravity belt thickener	T	Screw Press	No	B	Cake
E1	Rotary drum and gravity thickeners	M	Centrifuge	No	U	Cake
E2	Gravity thickeners	None	Belt Filter Press	No	U	Cake

¹ A – class A biosolids B – class B biosolids M – mesophilic digestion T – thermophilic digestion U-unclassified.² Solids produced were from a pilot plant which were subsequently recycled at the headworks of the facility producing A1. This represented approximately 2% of the A1 solids loading.

(Lakshminarasimman et al., 2021). Lakshminarasimman et al. (2021) also observed some modest reduction (10 to 38%) in some of the aerobic or anaerobic digestion trains, but reasons for these reductions were unclear with no consistent trends (Lakshminarasimman et al., 2021). Lazcano et al. (2020) documented the presence of total oxidizable precursors (TOP) in biosolids-based fertilizers, with approximately two to three times higher PFAAs after oxidation than present initially in the biosolids, indicating substantial transformation potential.

PFAS have been detected in surface and/or ground waters impacted by biosolids leachate, particularly in cases where PFAS industrial dischargers have caused unusually elevated levels of PFAS in land-applied biosolids (Boulanger et al., 2005; Lindstrom et al., 2011; Schultz et al., 2006; Sinclair and Kannan 2006; 2006; Xiao et al., 2012). Biosolids from WRRFs with no industrial inputs also have detectable PFAS concentrations and, when land applied, these PFAS migrate through the soil profile (Sepulvado et al., 2011), and have resulted in ng/L increases in PFAS levels in tile drainage water and shallow (2-m) groundwater (Gottschall et al., 2017). Long-term (12 years) sustained leaching of PFAS, primarily PFOS, following the land application of biosolids has been observed in a field site with land applied industrially-impacted

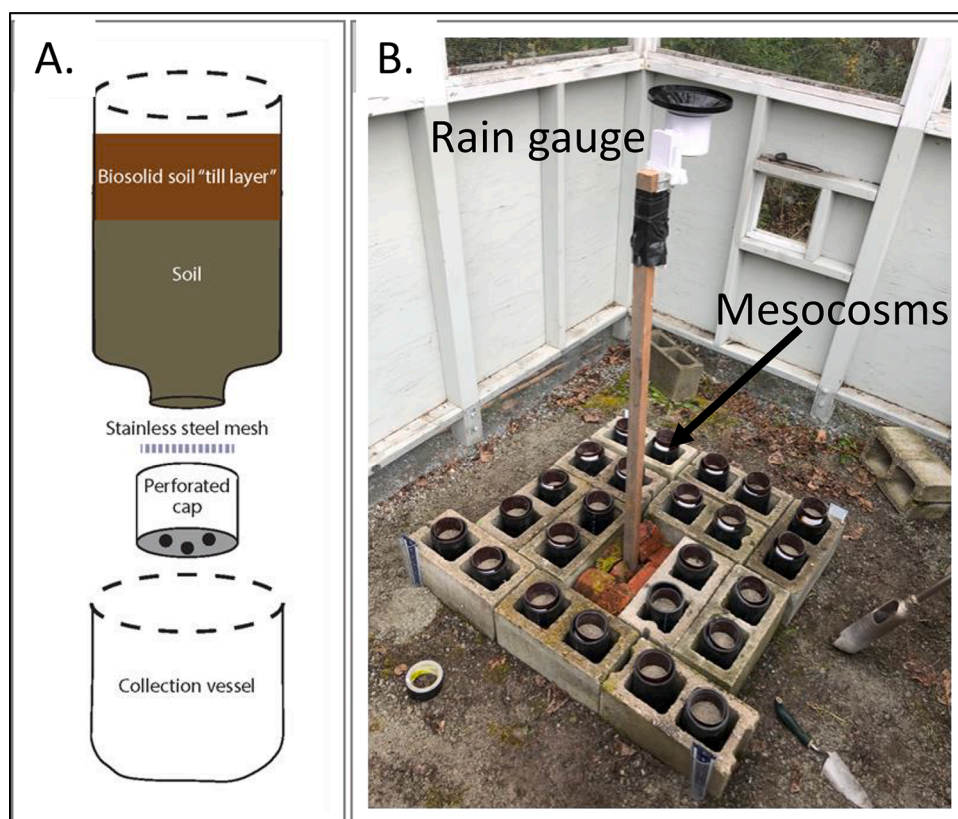
compost (Röhler et al., 2021). Ebrahimi et al. (2021) found that the relative release of PFAS from biosolids was impacted by protein content, organic carbon content, and lipid fraction.

While both PFAS composition and leaching from finished biosolids have been examined, quantifying directly the role of potential PFAA precursors on leaching from biosolids is lacking. The objective of this study was to determine the extent to which quantifiable polyfluorinated precursors present in finished biosolids contribute to overall PFAS leaching. Precursor composition and subsequent leaching among various biosolids with differing characteristics and solids processing are compared in a long-term outdoor leaching study.

2. Experimental

2.1. Collection of finished biosolids from WRRFs

Seven WRRFs at five public utilities across the United States participated in this study. Current average daily flows at these WRRFs ranged from less than 7 to 630 million gallons per day (MGD). The seven WRRFs were representative of various treatment processes ranging from

**Fig. 1.** Leaching mesocosm column construction (A) and outdoor deployment (B).

primary and secondary treatment to tertiary and advanced treatment. The facilities were from highly urbanized metropolitan areas, which included both domestic and industrial dischargers to the WRRFs. Tertiary wastewater treatment included nutrient removal of either phosphate or nitrogen and phosphate. Advanced wastewater treatment included granular media filtration, membrane filtration, pre-ozonation, and biological activated carbon (BAC). Disinfection included chlorine, ozone, or ultraviolet (UV) irradiation. The solids processing trains involved thickening, digestion, dewatering and/or drying. For the seven biosolids in this study, which were identified with the same IDs as the facilities from which they were obtained (IDs A1, A2, B1, C1, D1, E1, and E2), solids and liquid processing details are summarized in Table 1 and Table S1, respectively.

Sampling devices used for collection of biosolids that came in direct contact with the samples were triple rinsed with 99% American Chemical Society (ACS) grade methanol (Integra) and then triple rinsed with American Society for Testing and Materials (ASTM) Type I ultrapure water prior to sampling. Finished biosolids from each facility were collected in high-density polyethylene (HDPE) sample jars on a single day after the final step in processing (including any drying) via time-weighted composite grab samples over a six-hour period. Collected samples were shipped on ice overnight to the laboratory where they were subsequently homogenized, lyophilized, and subsampled for baseline characterization and PFAS analyses. Lyophilization was performed by initially freezing samples at -20°C for at least three hours before placing in a Labconco lyophilizer at approximately -50°C with an operational vacuum of 0.1 to 0.01 millibar. The duration of freeze-drying varied depending on the sample moisture content. Baseline biosolids characteristics determined on lyophilized samples are summarized in Table S2.

2.2. Column mesocosm leaching experiments

Outdoor column leaching experiments were performed to evaluate the impact of weathering and aging over a six-month period (April 2020–October 2020) with respect to PFAS leaching from the collected biosolids, thereby serving as a screening-level means to assess PFAS leaching from land-applied biosolids. Columns were constructed using 1-L amber HDPE Nalgene® bottles with a 4-inch inner diameter. All bottles were rinsed with methanol and ultrapure water prior to use. The bottle bottoms were removed, perforations were added into bottle cap, a stainless-steel mesh was affixed to the cap interior, and all cut edges were smoothed out by sanding. The bottom-less bottles were inverted and used as a collection vessel for leachate during the experiment. The experimental system is shown in Fig. 1A.

Sand used in the column experiments was obtained from Washington Rock Quarries (Orting, WA). Before use, the sand was rinsed with tap water and sieved using a #60 sieve (250 μm) to facilitate adequate drainage and removal of excess organic carbon (roots, leaves, debris). After sieving, the sand pH was 7.7 SU (1:1 sand in deionized water). The total organic carbon of the rinsed and sieved sand was 0.1%. PFAS analysis of the sand showed that all analytes were below the limit of quantification.

Approximately 500 g of sand were added into the inverted, capped bottle. On top of this sand layer, a mixed sand:biosolids layer was added consisting of 245 g sand and 5 g lyophilized biosolids (2% biosolids by mass in this simulated till layer, equating to 4.4 U.S. dry tons per acre). This biosolids addition is comparable to a typical biosolids addition rate for small grains agricultural land (2–5 dry tons per acre) (USEPA, 2020). The total depth of the porous media bed was 4 inches including the 1.3-inch till layer). All leaching columns were prepared in triplicate for each of the biosolids, resulting in 21 biosolids-amended columns. In addition, three controls comprised of only sand (i.e., no biosolids) were constructed using the same procedure but contained 750 g of sand without a “till” layer. These controls were intended to account for potential PFAS leaching from the sand, leaching from any components of

the experimental apparatus, and any PFAS collected from the natural rainwater.

The columns were transported to a community garden in Seattle, WA for deployment on April 1, 2020 and secured in a locked enclosure. The columns were placed within a four-inch HDPE pipe supported by a cinder block frame. An ECOWITT WH5360B digital rain gauge was erected in the center of the column plot to collect precipitation data (Fig. 1B). A total of five precipitation events over the course of the six-month column leaching experiment (April 1, 2020–October 17, 2020) yielded sufficient leachate collection (Figure S1) for PFAS and total organic carbon (TOC) analysis. Outdoor temperatures ranged from 2 to 36°C during the duration of the study. After each rain event, the total rainfall was recorded from the digital rain gauge and total leachate volume was measured. For each of the 5 leachate sampling events, all cumulative leachate was removed from the collection vessel. Subsamples of this collected leachate from each of the collection events were transferred into 50-mL polypropylene centrifuge tubes and stored at 4°C prior to analysis, thereby allowing for determination of the total quantity of PFAS that eluted from the columns. After collecting the final leachate samples, the biosolids layer for each column was removed, homogenized, and sub-sampled for PFAS analysis.

2.3. Analytical methods

Analyses for biosolids characterization (ammonia, calcium, % solids, organic carbon, and pH) were performed by a commercial laboratory (Soiltest Labs, Moses Lake, WA) using the standard methods cited in Table S2. PFAS extraction of solids followed previously described methods (Sepulvado et al., 2011; Dasu et al., 2013; Royer et al., 2015; Lazcano et al., 2020) with a few modifications as detailed in the Supplementary Materials along with details on the PFAS solid phase extraction method used for aqueous samples. PFAS were analyzed using a Shimadzu ultra-performance liquid chromatography (uPLC) system coupled to a Sciex 5600+ Triple Quadrupole Time-of-Flight/Mass Spectrometry (QToF/MS) or Shimadzu LC-3Q-MS/MS 8040, both in negative electrospray ionization mode (ESI). Suspect screening in positive ESI was also performed on one replicate of each biosolids but no additional PFAS were detected. Additional details regarding PFAS analysis, including semi-quantification of suspect analytes, are described in the Supplementary Materials. Table S3 lists the PFAS identified in this study. Table S4 provides the quality criteria used for the PFAS analysis. The total oxidizable precursor (TOP) assay was also performed on the extracts of the lyophilized biosolids to further evaluate the extent of potential precursors present using the method described by Houtz and Sedlak (2012) but modified for biosolids extracts (detailed in Supplementary Materials).

2.4. Statistical analysis for column leaching experiments

As described previously, triplicate sand-only control columns were deployed along with triplicate biosolids-amended columns to account for any PFAS not associated with leaching from the biosolids (i.e., background PFAS). PFAS concentrations in leachate from the biosolids-amended columns were corrected for background PFAS by subtracting the average PFAS concentrations in the leachate from the sand-only controls. To calculate 95% confidence intervals (CI) for these results (the difference between two means) Eq. (1) was employed:

$$\bar{x}_1 - \bar{x}_2 \pm z^* \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \quad (1)$$

where:

$\bar{x}_1$ = average PFAS concentration in biosolids-amended columns

$\bar{x}_2$ = average PFAS concentration in sand-only control columns

z^* = 1.96 for 95% confidence interval (CI)

σ_1^2 = standard deviation of PFAS concentration biosolids-amended

Table 2

Sum of PFAS quantified in collected biosolids. All concentrations are on a dry weight basis, and were determined following lyophilization. Values for the average and 95% confidence interval (CI) are shown.

Facility	Total PFAAs (µg/kg)		Total PFAS (µg/kg)	
	Average	95% CI	Average	95% CI
A1	38.2	1.46	323	14.1
A2	82.0	3.84	398	13.7
B1	32.8	0.969	1100	43.8
C1	59.2	1.75	600	21.7
D1	30.3	0.933	399	15.6
E1	25.4	0.617	365	11.3
E2	203	12.3	863	35.5

columns squared

σ_2^2 = standard deviation of PFAS concentration in sand-only controls squared

n_1 = number of observations for biosolid-amended columns

n_1 = number of observations for sand-only controls

3. Results and discussion

3.1. Biosolids characterization

Total concentration of 18 PFAAs and the total of all detected PFAS are summarized in Table 2 with individual PFAS concentrations provided in Table S5. Overall, 95% confidence intervals are relatively low (typically less than 5% of the average) indicating a high level of repeatability among the triplicate analyses. Total PFAS concentrations ranged from 323 ± 14.1 to 1100 ± 43.8 µg/kg (dry weight basis), which is comparable to the range of 4.2 to 910 µg/kg reported for Australian biosolids by Moodie et al. (2021), who monitored a similar suite of PFAS. Also, the range in total PFAAs is similar to those reported for 2014 commercial biosolids of 9.0 to 199 µg/kg (Lazcano et al., 2020). The relatively narrow range (i.e., within the same order of magnitude) among the facilities shown in Table 2 suggests that municipal wastewaters that are not highly impacted by industry likely carry a fairly

consistent typical PFAS load. This narrow range also suggests that the differences in the various wastewater and solids treatment processes (Table 1) among the facilities examined do not have substantial impacts on PFAS concentrations in finished biosolids.

PFAS in each of the biosolids on a fluorine molar basis notably shows that potential PFAA precursor compounds account for 77 to 97% of the total molar fluorine present (Fig. 2). Quantified precursor compounds were consistently dominated by the 5:3 and 7:3 fluorotelomer acids (FTAs) and the di-substituted polyfluorinated phosphate esters (e.g., 6:2 diPAP and 8:2 diPAP). Fluorotelomer acids may be intermediates from transformation of other PFAS (Dinglasan et al., 2004). DiPAPs are commonly used as coating agents in food-contact materials and found to migrate into foods (Begley et al., 2008), surface treatment and protectant applications (Buck et al., 2011), and personal care products and pharmaceuticals (Lee and Mabury 2011), thus their appearance in biosolids may stem from their widespread use and consumption. Moodie et al. (2021) also noted that diPAPs were the predominant PFAS detected in biosolids.

Interestingly, the long-chained sulfonates were prevalent over the short-chained sulfonates (91% long chain and 9% short chain for the perfluoroalkyl sulfonic acids (PFASs), on average, where long-chain is defined as greater than 6 carbons and short-chain is 6 or less carbons). This may be due to the increased hydrophobicity of long chain PFAS relative to short chain. However, for the perfluoroalkyl carboxylic acids (PFCAs), both long- and short-chain were equally represented (average of 55% and 45%, respectively).

The PFAS analyses performed on the biosolids interrogated a wide range of PFAS; however, it is recognized that many PFAS in the biosolids likely went undetected because they were either not extracted (e.g., side-chain polymers require stronger organic solvents to extract (Letcher et al., 2020)), MS detector parameters were not optimized for untargeted PFAS, or standards were not available. Therefore, the quantity of PFAA precursors present is likely underestimated.

The TOP assay, conducted to further characterize potential PFAA formation from extractable precursors, yielded higher total PFCAs on a fluorine molar basis than observed in the biosolids with increases ranging from 20% for A2 to a nearly 14-fold increase for A1 (Fig. 3).

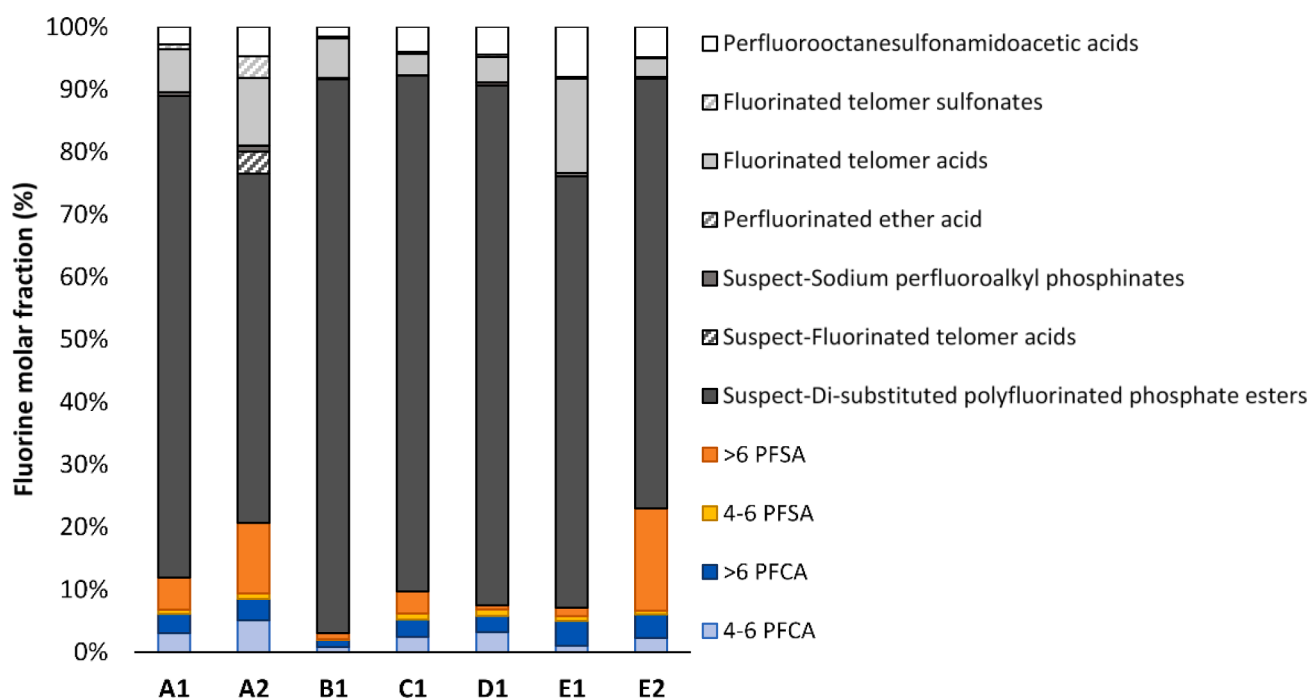


Fig. 2. PFAS distribution, on a fluorine basis, in the finished biosolids. 4–6 indicates a chain length of 4 to 6 carbons; >6 indicates a chain length of greater than 6 carbons. PFSA = perfluoroalkyl sulfonic acid. PFCA = perfluoroalkyl carboxylic acid.

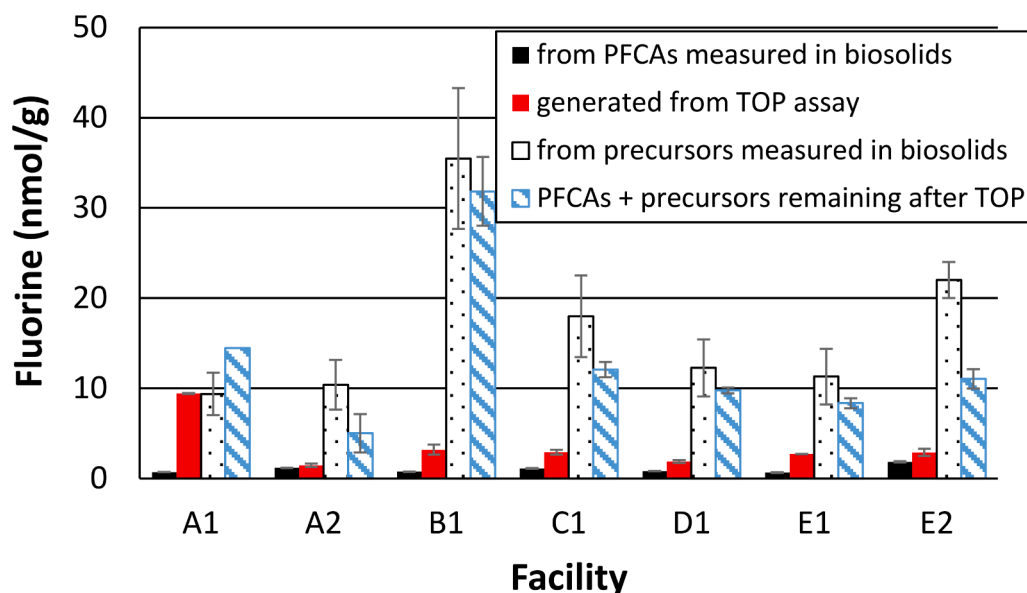


Fig. 3. Total fluorine from PFCAs measured in initial biosolids, measured after the TOP assay, estimated based on the potential conversion of precursors measured in the initial biosolids, and measured after the TOP assay with inclusion of persisting precursors. Averages of triplicate analyses with 95% confidence intervals are shown. Post-TOP assay results with inclusion of precursors were performed on duplicate samples only, with the exception of A1 where the analysis was performed on a single sample.

When the TOP assay results (based on PFCA generation) were compared to the fluorine contained in known and measured precursors (Tables 2 and S5), the TOP assay typically accounted for less than 25% of the fluorine in known precursors that were quantified directly. Thus, the TOP assay did not adequately account for the total known precursors quantified based on fluorine with the exception of A1. For A1, the TOP assay predicted a similar amount of fluorine as found in the quantified precursors in the initial biosolids sample.

One possible reason why the TOP assay failed to capture the majority of known precursors is that many of the identified precursors (e.g., diPAPs) may not have been fully transformed to PFCAs during the TOP assay, due to the chemical nature of these compounds and/or matrix issues associated with the biosolids extracts. To explore this possibility, the persistence of precursors following application of the TOP assay was determined. Results confirmed that substantial precursor mass persisted (Fig. 3), and that the TOP assay failed to convert all known precursors to PFCAs. The predominant precursors that persisted (accounting for >95% of the fluorine mass associated with these precursors) were 6:2 dPAP, 8:2 diPAP, and 5:3 fluorotelomer acid. In particular, the diPAPs showed minimal (<25%) transformation during the TOP assay. As shown in Fig. 3, the persistence of these precursors in addition to PFCA generation following the TOP assay accounted for 48 to 80% of the known fluorine mass initially present in the biosolids. The exception to these observations on precursor persistence following the TOP assay was again A1, where the fluorine balance following the TOP assay (with inclusion of precursors) exceeded that of the initial known fluorine mass in the biosolids. This exceedance could be the result of PFCA generation from precursors that were not identified in the biosolids prior to the TOP assay.

Even with accounting for the persistence of precursors following the TOP assay, precursor fluorine levels are significantly underpredicted in several of the biosolids. One possible explanation for this underprediction is based on a recent study that showed that the TOP assay resulted in formation of several ultra-short-chained PFCAs (e.g., trifluoroacetic acid (C2), and perfluoropropanoic acid (C3) that escape detection (Janda et al., 2019). PFCAs with chain lengths less than four are difficult to measure analytically due to significant matrix effects from salts generated during the TOP assay (potassium or sodium and sulfate from the oxidation of persulfate). The generation of ultrashort-chained PFCAs is especially important when the TOP assay is applied to samples with short-chain PFAS. For example, PFAS with C6 to C4 moieties, such as 4:2 fluorotelomer sulfonate and 6:2 fluorotelomer

sulfonate, may degrade to C2 and C3 products that are not typically analyzed. In the current study, approximately half of the PFCAs and PFASs detected in the biosolids were short chain. In any case, results shown herein suggest caution should be used when basing potential PFAA formation in biosolids solely on the TOP assay.

3.2. Leaching from biosolids

TOC concentrations measured in the leachate collected in the rainfall events from April until October 2020 decreased over time in all the biosolids columns from an average of 63 ± 35 mg/L for the first sampling event to 9.8 ± 2.4 mg/L for the last sampling event (Figure S2). The TOC in the effluent of the sand control column was relatively consistent at 7.5 ± 2.9 mg TOC/L. The total TOC released during the column experiment represented between only 1.3 to 2.9% of the total carbon mass initially present. TOC was not measured in the leachate from the first rainfall event because of limited leachate volume (only 1 pore volume of leachate was collected after the first rainfall event, as shown in Figure S1).

Measurable PFAS release from the sand controls was observed during the 6-month study (Figure S3). PFAS concentrations in the sand control leachate (Figure S3) were typically less than 40 ng/L for individual PFAS, with the exception of PFHxA for which release concentrations were approximately 100 ng/L through the first four pore volumes. PFAS were not detected in laboratory blank samples nor within the sand prior to the column experiments. The source of the PFHxA has not been determined; however, PFAS concentrations on the order of tens of ng/L have been observed in urban rainwater (Pike et al., 2021). PFAS concentrations in rainwater were not measured during the leaching study; however, given the urban environment of this study, PFHxA may have been present in the rainwater. The results for the biosolids-amended columns presented below were corrected for PFAS concentrations measured in the sand controls (i.e., PFAS concentrations in the controls were subtracted from the PFAS concentrations measured in the biosolid-amended columns).

PFAS concentrations observed in the leachate of the biosolids-amended columns include the PFAAs and four precursors (Figures S4 through S6 for PFCAs, PFASs, and precursors, respectively). Individual PFCAs and PFASs detected in the leachate generally ranged in the 10 s to 100 s of ng/L with PFCA concentrations being approximately double the PFSA concentrations. Precursor concentrations generally ranged between 10 and 100 ng/L. PFCA and PFSA releases from the columns were

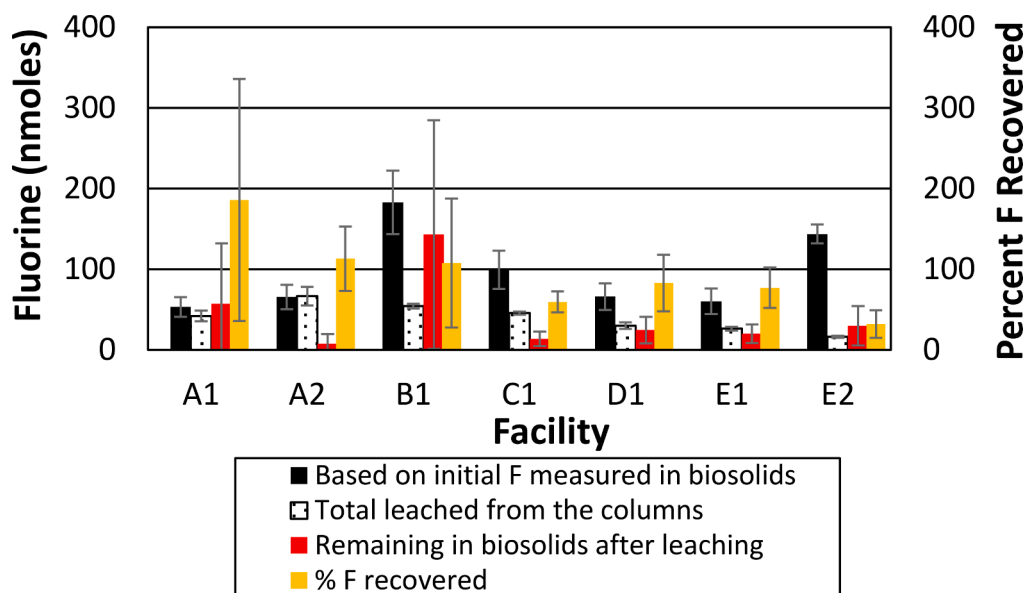


Fig. 4. Fluorine mass balance, considering fluorine present in the initially emplaced biosolids, the total fluorine leached from the columns over the 6-month study, and the fluorine remaining in the biosolids at the end of the 6-month leaching study. Averages of triplicates are shown. Error bars represent 95% confidence intervals. Also shown on the secondary y-axis are the percentages of fluorine recovered during the study (via collected leachate and fluorine remaining in the biosolids) compared to the initial fluorine present in the emplaced biosolids.

sustained for the duration of the six-month leaching study. Overall, no significant trends were apparent in either increasing or decreasing total PFAS leachate concentrations within the 6-month study (Table S6). This is consistent with observations from field-applied biosolids documented by others (Röhler et al., 2021), where sustained leaching occurred for up to several years.

3.3. PFAS and fluorine balances

Individual cumulative PFCAs and PFSAs leached from the columns relative to the PFAS initially present in the biosolids are shown in Figures S7 and S8, respectively. Values greater than one on the y-axis indicate that greater than 100% of the initial PFAA mass detected in the biosolids was observed in the leachate, as most notably seen for the PFCAs. Thus, the total mass of PFCAs desorbed into leachate far exceeded the initial PFCAs present in the biosolids. The presence of PFAA precursors and their subsequent transformation during the 6-month leaching experiment provides a plausible explanation for these results. Similar results, where increases in some PFCAs were observed over time, have been observed by others in biosolids field leaching studies (Venkatesan and Halden, 2014). The total mass of PFSAs leached from the columns relative to the initial mass in the emplaced biosolids also exceeded values of 1, but to a substantially lower extent than that of the PFCAs. Such behavior is consistent with previous observations that PFAA precursors are primarily biotransformed to PFCAs as end products (Lee et al., 2014; Harding-Marjanovic et al., 2015; D'Agostino and Mabury, 2017).

Total fluorine (nmoles) initially quantified in the biosolids, measured in leachate, and remaining in the aged mesocosm biosolids-sand mix post-leaching are summarized in Fig. 4, along with the % fluorine recovered (right axis) relative to the initial fluorine quantified in the biosolids emplaced in the mesocosms. A discussion of these calculations are provided in the Supplementary Materials. Fluorine molar balances show that the total fluorine eluted from the columns did not exceed the total fluorine initially present in the biosolids, despite the fact that the mass of individual PFCAs and PFSAs leached far exceeded what was initially present in the biosolids (Figures S7 and S8). These results indicate that the sustained leaching of PFAAs from the columns is in large part due to biotransformation of the precursors initially present in the biosolids. It is also noted that the absence of any precursors in the mesocosm control sand at the end of the 6-month experiment, as well as the absence of any precursors detected in the control effluents (with the

exception of trace levels of 6:2 fluorotelomer sulfonate, as shown in Figure S3), strongly suggests that the source of the precursors that were transformed were not from any precursors present in the rainwater, but rather from the biosolids.

For A1 and A2, the fluorine leached from the columns was nearly equal to the PFAS initially present in the biosolids. However, analysis of the biosolids post-leaching showed that some PFAS-associated fluorine was still present in the biosolids. One potential explanation for this discrepancy is that fluorine associated with unidentified PFAA precursors (e.g., low levels of cationic or zwitterionic PFAS not picked up in the ESI positive mode suspect screening) also may have been initially present in the biosolids and contributed to the formation of PFAAs, thus the fluorine initially present in the columns (Fig. 4) may be greater than what was quantified.

DiPAPs were the primary source of fluorine identified in the biosolids prior to leaching (Fig. 2) and the diPAPs decreased in the biosolids following the 6-month leaching study (differences between the black and red bars in Fig. 4 largely reflect the depletion of diPAPs in the biosolids). The diPAPs were also the only remaining PFAS identified in the biosolid-sand mix post leaching (Table S7), but were not observed in any of the leachate samples. Thus, it is likely that the biotransformation of diPAPs to primarily PFCAs controlled the overall fluorine leaching. Previous studies have shown that diPAPs in biosolids can biotransform to PFCAs, where the initial transformation step is hydrolysis of the phosphate ester bond (Lee et al., 2014).

The overall percent fluorine recoveries (Fig. 4), considering the relatively large error associated with the post-leaching biosolids sampling, were reasonably close to 100%. The exception was E2, where the fluorine mass recovered in the leachate and post-leaching biosolids was only 32% of the fluorine initially present in the biosolids; this may be due to some PFAS transforming to unquantified intermediates. Overall, the reasonable closure of these mass balances provide another line of evidence that the generated PFAAs were derived from precursors present in the biosolids, and not potentially unidentified precursors present in the rainwater.

The ratio of the fluorine mass leached from the columns relative to the fluorine initially present in the biosolids was not associated with a specific sludge process or type of biosolid produced. However, when comparing the overall fraction of fluorine released during the leaching study among the different biosolids, the fraction of fluorine released was linearly correlated to the organic carbon content of the biosolids (Fig. 5), where fractional removal of fluorine increased (or fluorine desorbed

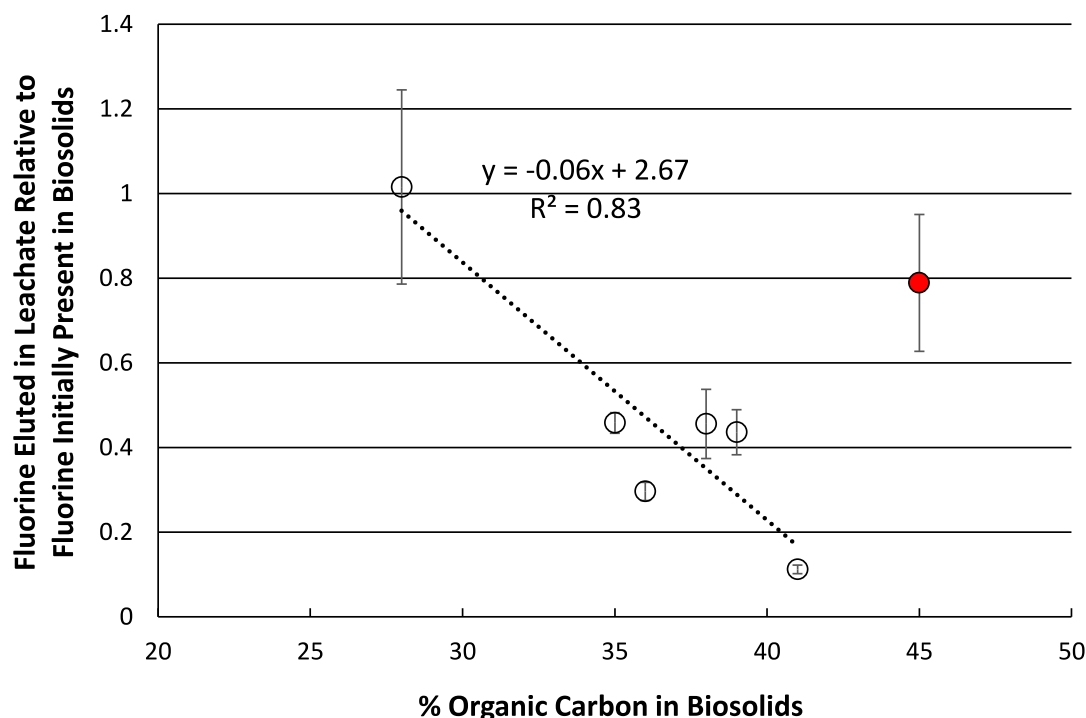


Fig. 5. Fluorine leached in the columns relative to fluorine initially present in the biosolids. The red-shaded datapoint is Facility A1, which was not used in the linear regression. Error bars represent the 95% confidence intervals.

increased) with decreasing organic carbon content. Interestingly, the one outlying datapoint in Fig. 5 was for A1. While it is unclear why A1 was an outlier, it is noted that the biosolids for A1 had the lowest pH (5.8), which may have had an impact on the sorption and/or transformation of PFAS.

5. Conclusions

For the finished biosolids examined in this study based on the 54 PFAS quantified, precursors (primarily diPAPs) accounted for the majority of the organic fluorine present. The outdoor mesocosm study showed that the PFAS and fluorine mass leached from the columns over a 6-month period could only be accounted for by considering precursor transformation. Both the standard analyte list (i.e., those for which analytical standards are available) and the TOP assay underestimated the PFAS mass in biosolids. While the results presented herein provide a strong line of evidence for the role of precursors in determining the sustained PFAS leaching from biosolids, additional studies are needed to assess the environmental impacts of such leaching following the land application of biosolids. The mesocosm experiment described herein does not address the fate of leachate in the soil, including further sorption, vadose zone migration, dilution, and other factors that impact PFAS concentrations in soil, groundwater, and other matrices.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.watres.2022.118405](https://doi.org/10.1016/j.watres.2022.118405).

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